

4.2 NONCLINICAL WRITTEN AND TABULATED SUMMARIES

AMLOZA (Amlodipine Besylate Tablets, USP) 5 mg / 10 mg — ANDA

Regulatory Basis

This application is submitted as an Abbreviated New Drug Application (ANDA) under Section 505(j) of the Federal Food, Drug, and Cosmetic Act. Consistent with 21 CFR 314.94(a)(8), an ANDA applicant is not required to submit new nonclinical (pharmacology, pharmacokinetic, or toxicology) study reports where the generic drug product does not raise nonclinical safety questions beyond those already addressed for the Reference Listed Drug (RLD). This application relies on FDA's prior finding of safety and effectiveness for the RLD, Norvasc® (amlodipine besylate) Tablets, NDA 019787 (Pfizer Inc.), approved in 1987.

Pharmacology

The primary pharmacodynamic profile of amlodipine — inhibition of calcium ion influx across cardiac and vascular smooth muscle cell membranes, resulting in coronary and peripheral vasodilation — is well established in the published pharmacological literature and is described in the approved RLD labeling (see Module 1.11, Section 12.1). No new primary or secondary pharmacodynamic studies were conducted, as AMLOZA contains the same active moiety, at the same strengths, via the same route of administration, as the RLD.

Pharmacokinetics

The nonclinical absorption, distribution, metabolism, and excretion (ADME) profile of amlodipine is extensively characterized in the RLD's approved labeling and in published literature accumulated over more than three decades of marketed use. No new nonclinical pharmacokinetic studies (e.g., animal ADME, mass balance, or drug-drug interaction studies) were conducted for this ANDA. The clinical pharmacokinetic bridge between AMLOZA and the RLD is instead established through the in vivo bioequivalence study summarized in Module 5.

Toxicology

No new nonclinical toxicology studies — including single- or repeat-dose toxicity, genotoxicity, carcinogenicity, or reproductive/developmental toxicity studies — were conducted for this application. The nonclinical toxicological profile of amlodipine besylate is considered adequately characterized by the nonclinical data reviewed and relied upon by FDA in its original approval of the RLD, and is reflected in the Warnings, Precautions, and Use in Specific Populations sections of the approved RLD labeling (see Module 1.11, Sections 5 and 8). No study reports, study numbers, NOAEL values, or histopathological findings are presented in this application, consistent with the ANDA pathway.

Excipients Assessment

All excipients used in the AMLOZA formulation (microcrystalline cellulose NF, anhydrous dibasic calcium phosphate USP, sodium starch glycolate Type A NF, magnesium stearate NF) are standard, compendial excipients with an established history of safe use in FDA-approved oral

solid dosage forms, including other approved amlodipine besylate products, as referenced in the FDA Inactive Ingredient Database. No novel excipients are introduced, and no new excipient-specific nonclinical safety data are required or presented.

Conclusion

Because AMLOZA (1) contains the same active ingredient, (2) is administered by the same route, (3) is presented in the same dosage form, and (4) is offered at the same strengths as the RLD, and because no novel excipients are introduced, no new nonclinical pharmacology, pharmacokinetic, or toxicology studies are required to support this ANDA under 21 CFR 314.94(a)(8). This application relies entirely on FDA's prior finding of safety and effectiveness for the RLD, supplemented by the bioequivalence data presented in Module 5, consistent with the generic drug approval pathway.

Portfolio Note: This summary demonstrates the correct ICH M4S/CTD structure and regulatory reasoning for Module 4 of a generic ANDA — explaining why nonclinical data are not generated, rather than fabricating study content that would not exist in a genuine 505(j) submission.

ACADEMIC PORTFOLIO EXERCISE — NOT FOR REGULATORY SUBMISSION